EP3120 - Statistical Mechanics

Kirit Makwana
Department of Physics, IIT Hyderabad

Fermi Gas

Fermi gas

- Electrons, protons, neutrons are all fermions
- Electrons in a metal can be thought of as a fermion gas
- These electrons are like free particles in a box with periodic boundary conditions with wavefunction

$$\psi(\boldsymbol{r}) = \frac{1}{\sqrt{V}} e^{i\boldsymbol{k}\cdot\boldsymbol{r}} \tag{1}$$

- box-normalized to volume V with energy

$$E\psi = -\frac{\hbar^2}{2m}\nabla^2\psi = \frac{\hbar^2 k^2}{2m}\psi \tag{2}$$

- The wavevector $\boldsymbol{k}$ is defined as

$$k_x = \frac{2\pi n_x}{L_x}; \quad k_y = \frac{2\pi n_y}{L_y}; \quad k_z = \frac{2\pi n_z}{L_z} \tag{3}$$

- The energy can be written as

$$E = \frac{\hbar^2}{2m}(k_x^2 + k_y^2 + k_z^2) \tag{4}$$

Energy levels in 3D box

- Each quantum energy level is labelled by (k_x, k_y, k_z) which can be thought of as a vector $\boldsymbol{k} = (k_x, k_y, k_z)$ with the energy

$$E = \frac{\hbar^2 k^2}{2m}; \quad k^2 = k_x^2 + k_y^2 + k_z^2 \tag{5}$$

- The average occupation number of a particular k state is

$$\langle n_k \rangle = (2S+1)\frac{1}{e^{\beta(\epsilon-\mu)} + 1} \tag{6}$$

- Here $2S+1$ is the degeneracy of a k state where S is the magnitude of the spin of the fermion
- This is because the fermions can have spin of $-S, -S+1, ..., S-1, S$ (in units of $\hbar$)
- For electrons that have $S = 1/2$, two states are possible $-1/2$ and $1/2$, which gives $2S+1 = 2$

Fermi level

- In the limit of temperature going to zero, the average occupation number of fermions becomes

$$\lim_{T\to 0} n_k = \lim_{T\to 0} (2S+1)\frac{1}{e^{\beta(\epsilon_k-\mu)+1}} \tag{7}$$

$$= \begin{cases} (2S+1) & \text{if } \epsilon_k < \mu \\ 0 & \text{if } \epsilon_k > \mu \end{cases} \tag{8}$$

- This can be written in terms of the step function Θ

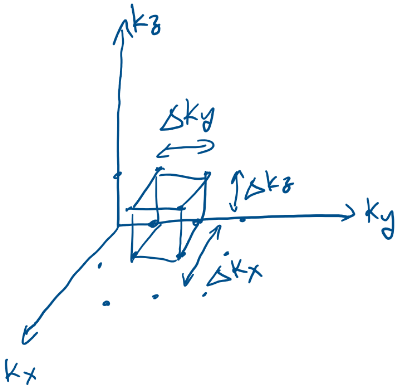
$$\lim_{T\to 0} n_k = (2S+1)\Theta(\mu-\epsilon_k) \tag{9}$$

- where

$$\Theta(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } x \geq 0 \end{cases} \tag{10}$$

Fermi level

- Each single particle k-state is a point in k-space



- The spacing between two points is

$$\Delta k_{x,y,z} = \frac{2\pi}{L_{x,y,z}} \tag{11}$$

Fermi level

- At $T = 0$ the sphere with radius

$$k_F^2 \leq \frac{2m\mu}{\hbar^2} \tag{12}$$

- will be completely filled and states outside this sphere will be completely empty
- The number of k-states in this sphere is

$$\begin{aligned} \text{No. of k-states} &= \frac{\text{Volume of sphere}}{\text{Volume of a single k-state}} && (13) \\ &= \frac{(4/3)\pi k_F^3}{(2\pi)^3 V^{-1}} && (14) \end{aligned}$$

- At each k-state there are $2S+1$ fermions, so the number of particles is

$$N = \frac{V(2S+1)}{(2\pi)^3}\frac{4\pi}{3}k_F^3 \tag{15}$$

Fermi surface

- This gives an expression for the Fermi surface/sphere

$$k_F = \left(\frac{6\pi^2 n}{2S+1}\right)^{1/3} \tag{16}$$

- where $n = N/V$ is the number density of electrons
- The corresponding Fermi energy is

$$\epsilon_F = \frac{\hbar^2 k_F^2}{2m} = \frac{\hbar^2}{2m}\left(\frac{6\pi^2 n}{2S+1}\right)^{2/3} \tag{17}$$

- Exercise: Calculate the energy of this Fermi gas at $T = 0$

Energy of Fermi gas

- The energy will be summed up over all the particles

$$E_0 = \sum_k (2S+1)\epsilon_k = \sum_k (2S+1)\frac{\hbar^2 k^2}{2m} \tag{18}$$

$$= (2S+1)\frac{V}{(2\pi)^3}\int_0^{k_F} 4\pi k^2 dk \frac{\hbar^2 k^2}{2m} \tag{19}$$

$$= (2S+1)\frac{V}{(2\pi)^3}\frac{\hbar^2}{2m}\frac{4\pi}{5}k_F^5 \tag{20}$$

- The average energy par particle will be

$$\frac{E_0}{N} = \frac{3}{5}\frac{\hbar^2 k_F^2}{2m} = \frac{3}{5}\epsilon_F \tag{21}$$

Pressure of Fermi gas

- Lets calculate the pressure that this fermi gas will exert
- Pressure will come from the momentum operator $-i\hbar\nabla$
- For the particle wavefunction

$$\psi(\boldsymbol{r}) = e^{i\boldsymbol{k}\cdot\boldsymbol{r}} \tag{22}$$

- The momentum operator gives

$$-i\hbar\nabla\psi = \hbar\boldsymbol{k}\psi \tag{23}$$

- This means that the expectation value of the momentum will be $\hbar\boldsymbol{k}$ for a given $\boldsymbol{k}$ state
- This also matches with the classical formula that $E = \frac{p^2}{2m}$

Pressure of Fermi gas

- The pressure can be evaluated as we did for a classical gas in kinetic theory

$$P = \int_{v_X>0} d\boldsymbol{p}(2mv_x)v_x f(\boldsymbol{p}) \tag{24}$$
$$= \int d\boldsymbol{p} m v_x^2 f(\boldsymbol{p}) \tag{25}$$

- $f(\boldsymbol{p})d\boldsymbol{p}$ is the number of particles per unit volume in region $d\boldsymbol{p}$
- In the $\boldsymbol{k}$-space now, this should be replaced by number of particles per unit volume in region $d\boldsymbol{k}$
- The number is given by

$$\frac{d\boldsymbol{k}}{(2\pi)^3} n_k (2S+1) \tag{26}$$

Pressure of fermi gas

- So the expression for the pressure becomes

$$P = \int d\mathbf{k} m v_x^2 n_k (2S+1) \tag{27}$$

- We can replace v_x^2 by $v^2/3$ and its relation to k is

$$v^2 = \frac{p^2}{m^2} = \frac{\hbar^2 k^2}{m^2} = \frac{2\epsilon_k}{m} \tag{28}$$

- Substituting the expression for the occupation number of fermions

$$P = \frac{2(2S+1)}{3(2\pi)^3} \int d\mathbf{k} \frac{\epsilon_k}{z^{-1} e^{\beta\epsilon_k} + 1} \tag{29}$$

- Make the substitution

$$x^2 = \beta\epsilon_k \Rightarrow k = \sqrt{\frac{2mk_B T}{\hbar^2}} x = \frac{\sqrt{4\pi}}{\lambda} x \tag{30}$$

- Here λ is the thermal De Broglie wavelength

Pressure of fermi gas

- Making the subsitution gives the result

$$\frac{P}{k_B T} = (2S+1)\frac{1}{\lambda^3}\frac{8}{3\sqrt{\pi}}\int_0^\infty dx \frac{x^4}{z^{-1}e^{x^2}+1} \tag{31}$$

- Here we have integrated over a spherical shell of volume $4\pi k^2 dk$ assuming that the distribution is spherically symmetric
- This is valid for both fermions and bosons (for bosons replace + sign with minus sign)
- This integral gives the result for pressure at arbitrary temperature-it can be evaluated numerically
- The energy is given by

$$U = \sum_k (2S+1)\frac{\epsilon_k}{z^{-1}e^{-\beta\epsilon_k}+1} = V(2S+1)\int d\boldsymbol{k}\frac{1}{(2\pi)^3}\frac{\epsilon_k}{z^{-1}e^{\beta\epsilon_k}+1} \tag{32}$$

Internal energy

- Comparing the expressions for the energy and pressure we get the result

$$PV = \frac{2}{3}U \tag{33}$$

- This is a general relation that works at any temperature
- Since we have already calculated the energy at absolute zero, we can evaluate the pressure at absolute zero

$$\begin{aligned} P_0 &= \frac{2}{3}\frac{E_0}{V} \qquad (34) \\ &= \frac{2}{5}n\epsilon_F \qquad (35) \end{aligned}$$

- Unlike in a classical ideal gas where pressure drops to zero at absolute zero, a fermi gas shows a finite pressure
- This arises because there is only one k-state with zero momentum, which gets filled and so other particles have to occupy higher momentum states

Fermi temperature

- Consider a metal with number density of electrons 10^{22}cm^{-3}
- H.W.: It can be calculated that the pressure this gas exerts at absolute zero is around 10^4atm
- This is also called as the zero-point pressure
- The Fermi temperature is defined as

$$\epsilon_F = k_B T_F \tag{36}$$

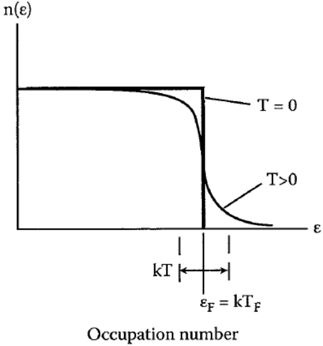
- At slightly higher temperature, the particle occupation number starts deviating from the step function

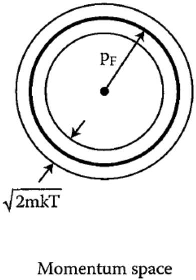
$$n_i = \frac{g_i}{e^{\beta(\epsilon_i - \mu)} + 1} \tag{37}$$

$$\approx g_i \begin{cases} e^{-\beta(\epsilon_i - \mu)} \text{ if } \epsilon_i > \mu \\ 1 - e^{\beta(\epsilon_i - \mu)} \text{ if } \epsilon_i < \mu \end{cases} \tag{38}$$

Finite temperature

- The occupation numbers will then behave as





- Some electrons will be transferred from below the fermi level to above the fermi level
- This will create vacancies below the fermi level called as holes
- In the momentum space it will be seen as particles just below the Fermi sphere moving to just outside the Fermi sphere

Finite temperature

- For low temperatures $T \ll T_F$ the distribution deviates from the step function only in a narrow region of width around $k_B T$ near the fermi surface $\epsilon = \epsilon_F$
- For $T \gg T_F$ the distribution becomes very broad and the Fermi surface disappears - the distribution becomes very close to Maxwell-Boltzmann distribution
- Electrons in a metal typically have $\epsilon_F \approx 2eV$ which corresponds to a Fermi temperature $T_F \approx 2 \times 10^4$K
- What is the fraction of electrons excited above Fermi level at room temperature?
- That is the area under the curve above Fermi level to area of rest of the curve

$$\text{Fraction} = \frac{\int_{\epsilon_F}^{\infty} e^{-\beta(\epsilon_i - \mu)} d\epsilon}{\epsilon_F} = \frac{e^{-\beta\epsilon_F}}{\beta\epsilon_F} \approx \frac{T}{T_F} \tag{39}$$

Heat capacity

- Since room temperature $T \approx 0.015 T_F$ it means that only 1.5% of electrons are excited above the Fermi level
- As each excited electron has increase in energy of around $k_B \Delta T$, the energy input is

$$Q \approx N\left(\frac{T}{T_F}\right) k_B \Delta T \tag{40}$$

- This gives a specific heat capacity of

$$c \approx k_B \frac{T}{T_F} \tag{41}$$

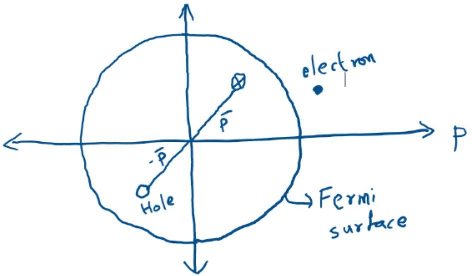
- Before the advent of quantum statistics, it was thought that heat capacity of electrons in solids should be similar to that of an ideal gas
- In an ideal gas, each degree of freedom contributes $k_B/2$ to the heat capacity, so it should be of the order of k_B

Heat Capacity

- But experimental measurements showed that the heat capacity of electrons in solids is negligible - instead, the heat capacity mostly comes from the lattice vibrations in a solid
- The result from quantum statistics explained the experimental observation since $T \ll T_F$ and so the specific heat capacity at around room temperature is $\ll k_B$
- The reason is that at room temperature very little energy is expended in exciting the electrons above Fermi level
- A careful derivation of the heat capacity at low temperatures gives the correct analytical behavior of the heat capacity of solids
- At higher temperatures comparable to the Fermi temperature, electron heat capacity rises and becomes close to $(3/2)k_B$
- The real contribution to the heat capacity of solids comes from phonons - lattice vibrations
- These can be treated as bosonic harmonic oscillators and they give the correct value for heat capacity of solids

Particles and holes

- At lower than Fermi temperatures, where only small fraction of electrons are excited above the Fermi level, there is an alternative way to describe the system
- Consider an electron at momentum $\boldsymbol{p}$ within the Fermi sphere gets energized out of it



- This system can be treated as a collection of electrons as they are, i.e., all electrons in the Fermi sphere except one, and one electron outside of it

Particles and holes

- Alternatively, one can imagine this system as a fully-filled Fermi sphere, one electron outside the Fermi sphere, and a hole
- In order to maintain charge, momentum, and energy the hole should have

$$q_h = -q_e \tag{42}$$
$$\boldsymbol{p}_h = -\boldsymbol{p} \tag{43}$$
$$\epsilon_h = -\epsilon \tag{44}$$

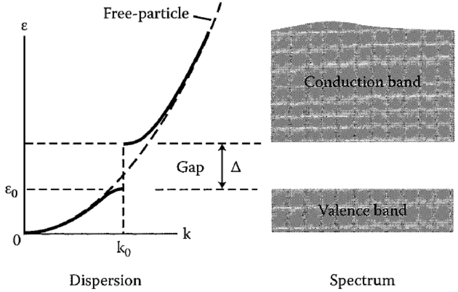
- Since the occupation number of fermions is 0 or 1, the occupation number of hole is

$$n_h = 1 - n_k = 1 - \frac{1}{e^{\beta(\epsilon-\mu)} + 1} = \frac{1}{e^{\beta(\mu-\epsilon)} + 1} \tag{45}$$

- In the case where the fraction of holes is less, the fully occupied Fermi sphere just acts as a background and the system can be treated simply in terms of electron-hole pairs

Electrons in solids

- Electrons moving freely in a box is a highly simplified model for electrons in a solid
- A more accurate description is that of electrons moving in a periodic potential setup by the lattice of ions
- Solving for the Schrodinger equation in a periodic potential gives a slightly different form of energy levels



- The free particle dispersion relation $\epsilon = \hbar^2 k^2/2m$ splits into bands

Electrons in solids

- The dispersion curve at the edges of the conduction and valence bands can be approximated by parabola
- The conduction band energy is

$$\epsilon_c = \epsilon_0 + \Delta + \frac{p^2}{2m_c} \tag{46}$$

- where Δ is the band gap and m_c is some effective mass
- The valence band energy is

$$\epsilon_v = \epsilon_0 - \frac{p^2}{2m_v} \tag{47}$$

- where Δ is the band gap and m_c, m_v are some effective mass, and $\boldsymbol{p}$ is some generalized momentum, $p = \hbar(k - k_0)$
- A natural semiconductor is a material which at absolute zero has completely filled valence band and fully empty conduction band

Electrons in semiconductor

- As the temperature as increases, holes are created in the valence band and particles occupy the conduction band
- The number densities of particles and holes will be

$$n_{part} = 2\int \frac{d\boldsymbol{p}}{h^3} \frac{1}{e^{\beta(\epsilon_c - \mu)} + 1} \tag{48}$$

$$n_{hole} = 2\int \frac{d\boldsymbol{p}}{h^3} \frac{1}{e^{\beta(\mu - \epsilon_v)} + 1} \tag{49}$$

- Assume that $\beta(\epsilon_c - \mu) \gg 1$ and $\beta(\mu - \epsilon_v) \gg 1$ then the exponential terms dominate

$$n_{part} \approx 2\int \frac{d\boldsymbol{p}}{h^3} e^{-\beta[\epsilon_0 + \Delta - \mu]} e^{-\beta p^2/2m_c} \tag{50}$$

$$= 2e^{-\beta[\epsilon_0 + \Delta - \mu]} \left(\frac{m_c k_B T}{2\pi\hbar^2}\right)^{3/2} \tag{51}$$

Chemical potential

- Similarly, the hole density will be

$$n_{hole} \approx 2 \int \frac{d\boldsymbol{p}}{h^3} e^{-\beta(\mu - \epsilon_v)} \tag{52}$$

$$= 2e^{-\beta(\mu - \epsilon_0)} \left(\frac{m_v k_B T}{2\pi\hbar^2} \right)^{3/2} \tag{53}$$

- Overall electrical neutrality requires that

$$n_{part} = n_{hole} \tag{54}$$

- This gives a relation for the chemical potential

$$\mu = \epsilon_0 + \frac{\Delta}{2} + \frac{3}{4} k_B T \ln \left(\frac{m_v}{m_c} \right) \tag{55}$$

- At absolute zero, the fermi energy becomes $\epsilon_0 + \Delta/2$

Electrons in semiconductors

- Substituting back in the expression for the electron and hole density gives the

$$n_{part} = n_{hole} \approx 2e^{-\beta\Delta/2}\left(\frac{\sqrt{m_c m_v}k_B T}{2\pi\hbar^2}\right)^{3/2} \tag{56}$$

- This is the carrier density. Take typical values $\Delta/(k_B T) = 0.7$, $m_c = m_v = m_e$
- This gives a carrier density of $1.6 \times 10^{13}\mathrm{cm}^{-3}$
- This is much smaller that the number density of electrons in a metal - so semiconductor is a poor conductor at room temperature

Statistical equilibrium of white dwarf stars

- The first application of Fermi statistics was in astrophysics
- White dwarf stars are that in which Hydrogen has burned up into Helium
- Their content is Helium and electrons
- The mass is roughly 10^{33}g of Helium, packed in a sphere of density $10^7 \mathrm{gcm}^{-3}$ at a central temperature of 10^7K
- There are N number of electrons with $N/2$ number of Helium nuclei (with mass $4m_p$)
- The total mass of the white dwarf star becomes $M \approx (N/2)4m_p = 2Nm_p$
- The electron density is

$$n = \frac{N}{V} = \frac{M/2m_p}{M/\rho} = \frac{\rho}{2m_p} \tag{57}$$

Fermi level

- From this we can estimate a typical number density of electrons in the white dwarf

$$n \approx 10^7/(2 \times 10^{-24}) \approx 10^{30} cm^{-3} \tag{58}$$

- The Fermi momentum of this electron gas will be

$$p_F = \hbar k_F = h\left(\frac{3n}{8\pi}\right)^{1/3} \approx 10^{-17} gcms^{-1} \tag{59}$$

- And the Fermi energy and temperature are

$$\epsilon_F = p_F^2/2m = 10^5 eV; \quad T_F \sim 10^9 K \tag{60}$$

- The electrons are relativistic because the Fermi energy is close to their rest mass energy
- The electron gas is in the degenerate limit because $T \ll T_F$

Pressure in white dwarf

- The Helium nuclei do not contribute significantly to the degeneracy pressure, instead it is supported mostly by the electron degeneracy pressure
- Lets evaluate the internal energy of this relativistic electron Fermi gas
- Relativistic energy is given as

$$\epsilon_p = \sqrt{p^2c^2 + m^2c^4} \tag{61}$$

- The zero-point energy will be given by

$$E_0 = \frac{2V}{h^3}\int_0^{p_F} dp 4\pi p^2 \sqrt{(pc)^2 + (m_ec^2)^2} \tag{62}$$

$$= \frac{2V}{h^3} 4\pi m_e c^2 \int_0^{p_F} dp p^2 \sqrt{1 + \left(\frac{p}{m_e c}\right)^2} \tag{63}$$

- Substitute $x = p/m_e c$

Pressure in white dwarf

- Then we get

$$E_0 = \frac{2V}{h^3} 4\pi m_e^4 c^5 \int_0^{x_F} dx x^2 \sqrt{1+x^2} \tag{64}$$

- Here we can define

$$f(x_F) = \int_0^{x_F} dx x^2 \sqrt{1+x^2} \tag{65}$$

- In the limit of $x_F \ll 1$ we can Taylor expand the square root to get

$$f(x_F) \approx \frac{x_F^3}{3} \left(1 + \frac{3}{10} x_F^2 + \ldots\right) \tag{66}$$

Pressure in white dwarf

- When $x_F \gg 1$, we can pull x^2 out of the square root

$$f(x_F) = \int_0^{x_F} dx x^3 \left(1 + \frac{1}{2x^2}\right) \tag{67}$$

$$\approx \frac{x_F^4}{4}\left(1 + \frac{1}{x_F^2}\right) \tag{68}$$

- x_F can be written as

$$x_F = \frac{p_F}{m_e c} = \frac{h}{m_e c}\left(\frac{3N}{8\pi V}\right)^{1/3} \tag{69}$$

$$= \frac{\hbar}{m_e c R}\left(\frac{9\pi M}{8m_p}\right)^{1/3} \tag{70}$$

- Define a normalized mass and radius

$$\bar{M} \equiv \frac{9\pi M}{8m_p} \quad and \quad \bar{R} = \frac{R}{\hbar/m_e c} \tag{71}$$

Pressure in white dwarf

- From first law of thermodynamics, if the volume of the star changes adiabatically by dV then the change in internal energy should be given as

$$dE_0 = -P_0 dV \Rightarrow P_0 = -\frac{\partial E_0}{\partial V} \tag{72}$$

- This gives

$$P_0 = -\frac{m_e^4 c^5}{\pi^2 \hbar^3}\left[f(x_F) + Vf'(x_F)\frac{\partial x_F}{\partial V}\right] \tag{73}$$

$$= -\frac{m_e^4 c^5}{\pi^2 \hbar^3}\left[f(x_F) - Vx_F^2\sqrt{(1+x_F^2)}\frac{x_F}{3V}\right] \tag{74}$$

$$= \frac{m_e^4 c^5}{\pi^2 \hbar^3}\left[\frac{x_F^3}{3}\sqrt{(1+x_F^2)} - f(x_F)\right] \tag{75}$$

Equilibrium of white dwarf

- In the non-relativistic limit $x_F \ll 1$ this becomes

$$P_0 \approx \left(\frac{m_e^4 c^5}{15\pi^2\hbar^3}\right) x_F^5 = \frac{4}{5} K \frac{\bar{M}^{5/3}}{\bar{R}^5} \tag{76}$$

- In the ultra-relativistic limit $x_F \gg 1$

$$P_0 \approx \left(\frac{m_e^4 c^5}{12\pi^2\hbar^3}\right) (x_F^4 - x_F^2) \tag{77}$$

$$= K\left(\frac{\bar{M}^{4/3}}{\bar{R}^4} - \frac{\bar{M}^{2/3}}{\bar{R}^2}\right) \tag{78}$$

- where

$$K = \frac{m_e c^2}{12\pi^2}\left(\frac{m_e c}{\hbar}\right)^3 \tag{79}$$

- Lets obtain a relation between the pressure, mass, and radius of the white dwarf star

Equilibrium of white dwarf

- The work done to compress the star against its Fermi degeneracy pressure is

$$dW = -P_0 4\pi r^2 dr \tag{80}$$

- The total work done to bring it in the equilibrium state is therefore

$$W = -\int_{\infty}^{R} P_0 4\pi r^2 dr = \int_{R}^{\infty} P_0 4\pi r^2 dr \tag{81}$$

- This work done will be stored in the form of gravitational potential energy
- The gravitational potential energy will be of the form (by dimensional analysis)

$$U_G = \alpha \frac{GM^2}{R} \tag{82}$$

- where α is some dimensionless number depending on the distribution of density inside the star
- This α will be of order unity

Equilibrium of white dwarf

- Equating the work done by gravity against pressure we get

$$\int_R^\infty P_0 4\pi r^2 dr = \alpha \frac{GM^2}{R} \tag{83}$$

- Differentiating w.r.t. R we get

$$-P_0 4\pi R^2 = -\alpha \frac{GM^2}{R^2} \tag{84}$$

$$\Rightarrow P_0 = \frac{\alpha GM^2}{4\pi R^4} \tag{85}$$

$$= \frac{\alpha}{4\pi} G \left(\frac{8m_p}{9\pi}\right)^2 \left(\frac{m_e c}{\hbar}\right)^4 \frac{\bar{M}^2}{\bar{R}^4} \tag{86}$$

- Now by equating the expressions for pressure already obtained in the non-relativistic and ultra-relativistic limits, we can obtain a relation between the mass and radius of the white dwarf

Mass-Radius relationship

- Define one more constant K'

$$K' \equiv \frac{\alpha}{4\pi} G \left(\frac{8 m_p}{9\pi} \right)^2 \left(\frac{m_e c}{\hbar} \right)^4 \tag{87}$$

- Then in the non-relativistic limit, equating the two expressions for pressure gives

$$\frac{4}{5} K \frac{\bar{M}^{5/3}}{\bar{R}^5} = K' \frac{\bar{M}^2}{\bar{R}^4} \tag{88}$$

$$\Rightarrow \bar{M}^{1/3} \bar{R} = \frac{4}{5} \frac{K}{K'} \tag{89}$$

- This is the limit where Fermi energy is non-relativisitic, i.e., density is low
- In this case,as the mass of the star increases, its radius decreases

Mass-radius relationship

- Equating the expression for the pressure in the ultrarelativistic limit gives

$$K\left(\frac{\bar{M}^{4/3}}{\bar{R}^4} - \frac{\bar{M}^{2/3}}{\bar{R}^2}\right) = K'\frac{\bar{M}^2}{\bar{R}^4} \tag{90}$$

$$\Rightarrow \bar{R}^2 = \bar{M}^{2/3}\left(1 - \frac{K'}{K}\bar{M}^{2/3}\right) \tag{91}$$

- Define

$$\bar{M}_0^{2/3} \equiv \frac{K}{K'} \tag{92}$$

$$\bar{R} = \bar{M}^{1/3}\sqrt{1 - \left(\frac{\bar{M}}{\bar{M}_0}\right)^{2/3}} \tag{93}$$

- The mass M_0 can be calulcated (taking $\alpha \approx 1$)

$$M_0 = \frac{8m_p}{9\pi}\left(\frac{27\pi}{64\alpha}\right)^{3/2}\left(\frac{\hbar c}{Gm_p^2}\right)^{3/2} \approx 10^{33}g \tag{94}$$

Chandrasekhar mass limit

- This formula applies in the high density ultra-relativistic limit
- The mass M_0 is close to the solar mass $M_\odot$
- A more detailed calculation (by S. Chandrasekhar, Physics Nobel prize 1983) shows a relation with

$$M_0 = 1.4 M_\odot \tag{95}$$

- This is called as the Chandrasekhar mass limit
- If a white dwarf has a mass more than $1.4M_\odot$ then this means that its radius will collapse to zero - implying that the Fermi degeneracy pressure cannot balance the gravitational force
- Stars with mass more than the Chandrasekhar mass limit collapse to form blackholes